IEEE Open Journal of Intelligent Transportation Systems

Received XX Month, XXXX; revised XX Month, XXXX; accepted XX Month, XXXX; Date of publication XX Month, XXXX; date of current version XX Month, XXXX.

Digital Object Identifier 10.1109/OJITS.2026.XXXXXXX

EgoPed: Ego-Conditioned Pedestrian Body Motion Generation via Latent Diffusion with Full-Sequence Ego Conditioning

ERIK SCHÜTZ[1], MOHAN RAMESH[1], AND FABIAN B. FLOHR[1]

[1]Intelligent Vehicles Lab, Munich University of Applied Sciences, Munich, Germany

CORRESPONDING AUTHOR: Erik Schütz (e-mail: erik.schuetz@hm.edu).

ABSTRACT We introduce EgoPed, a system that generates 3D full-body pedestrian motion conditioned on vehicle ego trajectories from autonomous driving data. Existing pedestrian motion synthesis relies on text descriptions or action labels, ignoring the driving context that shapes how a person walks, stops, or reacts. We condition a Motion Latent Diffusion model on ego odometry by exposing the denoiser to all 196 per-timestep ego tokens rather than a pooled summary, through a frozen, alignment-pretrained ego encoder. On a benchmarked consisting of combined nuScenes, Waymo and self-recorded data, full-sequence conditioning improves FID by **40%** over the strongest pooled baseline (5.620 $\rightarrow$ 3.392), at equal denoiser capacity and against that baseline's best seed. We further show that FID alone cannot detect whether the condition is used as separately trained unconditional models reach 3.24, and 1.43 under interaction-aware sampling, while remaining ego-blind and introduce per-condition tests that can. On a scene-disjoint held-out split, these tests confirm that full-sequence conditioning uses the vehicle: it leads on trajectory error (ADE 2.46 $\rightarrow$ 2.36 m) and on the coupling between generated stopping decisions and the vehicle (0.28 $\rightarrow$ 0.33), both significant under a paired bootstrap. Ablations show the frozen encoder is critical and the attention mechanism is not. Furthermore, interaction-aware sampling improves the marginal (FID 2.67–2.88 over two seeds) but not per-condition fidelity. Finally, a reference-free audit of the pose labels which uses one canonical skeleton for every pedestrian, and feet that essentially never plant, bounds what FID can mean on this benchmark. The code along with trained weights and the used dataset will be made publicly available.

INDEX TERMS Pedestrian behaviour modelling, human motion synthesis, denoising diffusion models, autonomous driving, vehicle–pedestrian interaction, generative model evaluation, motion capture data quality.

I. INTRODUCTION

GENERATING realistic pedestrian body motion is a critical capability for autonomous vehicle (AV) simulation. Simulation pipelines require diverse, plausible pedestrian behaviour to stress-test perception and planning before deployment, and especially for the long tail of close-range encounters that are unsafe or impractical to collect on the road. The intelligent-transportation literature has long established that this behaviour is coupled to the vehicle: the ego vehicle's speed, trajectory and proximity all influence whether a pedestrian crosses, waits, accelerates or turns [1], and benchmarks such as JAAD [2] and PIE [3] annotate exactly that dependence. Those benchmarks, however, predict discrete events, like a crossing intention, and the models built on them output a label or a ground-plane trajectory [4]–[6], not an articulated body.

Full-body human motion synthesis has meanwhile advanced rapidly, driven by text-conditioned diffusion models [7]–[10]. These systems are context-agnostic in precisely the way an AV simulator cannot afford: a pedestrian crossing a street is generated identically whether a vehicle is approaching at 50 km/h or standing still. The *ego-conditioned* setting that joins both parts by generating full 3D body motion conditioned on the vehicle's own trajectory is largely unexplored. Existing work either generates 2D trajectory distributions only [11], or uses ego context as scene conditioning without 3D body synthesis [12], while spatially controlled motion models [13], [14] constrain the generated person directly

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

rather than conditioning on a second, independently moving agent.

This paper addresses ego-conditioned *full-body* pedestrian motion synthesis. We build on Motion Latent Diffusion (MLD) [7], which compresses 263D HumanML3D body motion into a compact latent space and denoises within it, but replace the text conditioning with vehicle ego trajectory. The key architectural question is: *how should the ego trajectory be injected into the denoiser?*

We study two families of conditioning mechanisms:

- **Pooled conditioning**: The ego trajectory is encoded by a transformer encoder and mean-pooled to a single 256D token, which is concatenated alongside the time embedding in the denoiser's self-attention. This gives the denoiser a coarse summary of the ego trajectory but no temporal resolution.
- **Full-sequence conditioning**: The ego trajectory is encoded to a full 196-token sequence, which is concatenated with the latent tokens and jointly self-attended at each denoising step. This exposes every denoising step to the full ego trajectory, enabling temporally fine-grained conditioning.

Our main finding is that full-sequence conditioning dramatically outperforms pooled conditioning. EgoPed achieves **FID = 3.392** versus the pooled baseline's 5.620 under a unified evaluation pipeline constituting a 40% improvement with the same denoiser capacity, whose best checkpoint occurs *earlier* in training (epoch 3399 vs. 4399), so the gain does not come from additional compute. Adding interaction-aware training-data sampling improves our best system further to FID = 2.880. The full-sequence denoiser can attend to specific moments in the ego trajectory rather than a coarse average, producing motion that more closely follows the ground truth distribution.

A secondary but equally important finding concerns the ego encoder's training regime. EgoPed uses a *frozen* ego encoder which is pretrained to align mean-pooled ego trajectories with VAE motion latents, then fixed during diffusion training. We ablate this choice by unfreezing the encoder which allows it to co-adapt with the denoiser. This approach consistently *underperforms* EgoPed on the distributional metrics at every evaluated checkpoint: FID worsens by 50%, R-Precision@1 drops by 36%, and MultiModality collapses by 34%. We show that this occurs because the unfrozen encoder specializes to discriminative, per-condition representations that over-constrain generation diversity. This is consistent with the frozen-encoder paradigm in image generation (frozen CLIP in Stable Diffusion [15], adapters on frozen backbones in IP-Adapter [16] and ControlNet [17]).

Our contributions are:

1) **Ego-conditioned body motion.** The first model to generate full 3D pedestrian body motion conditioned on the ego-vehicle trajectory, trained on AVA, nuScenes [18], and Waymo [19].
2) **Full-sequence conditioning.** Attending to all 196 ego tokens instead of one pooled token improves FID by 40% at equal capacity and lowers per-condition trajectory error.
3) **Conditioning-aware evaluation.** We show that FID cannot detect whether the condition is used — an ego-blind model reaches the best FID in our study — and introduce per-condition tests that can.
4) **Design guidance.** Freezing the pretrained ego encoder preserves generative diversity; interaction-aware sampling improves FID but not per-condition fidelity.
5) **Pose-label audit.** A reference-free audit shows that the pseudo-ground-truth root trajectories are reliable, while the body poses are a sliding canonical template.

II. RELATED WORK

A. PEDESTRIAN BEHAVIOUR MODELLING FOR AUTOMATED DRIVING

Modelling how pedestrians behave around vehicles is a long-standing concern in intelligent transportation. Camara *et al.* survey the area in two parts, separating low-level sensing, detection and tracking [20] from high-level models of pedestrian decision-making and vehicle–pedestrian interaction [1]. The dominant experimental paradigm is video-based and predicts discrete events. JAAD [2] annotates crossing behaviour together with its situational context while PIE [3] adds large-scale intention labels alongside ego-vehicle sensor readings, making the vehicle state an explicit input. Kotseruba *et al.* [21] re-evaluate pedestrian action prediction under a common protocol and find that reported gains are strongly sensitive to evaluation design and to the base rate of the predicted event.

We share the domain but change the output. Rather than classifying whether a pedestrian will cross, we synthesise the full articulated body motion conditioned on the vehicle trajectory. The protocol lesson of [21] transfers directly, and is one motivation for the per-condition controls in Sections IV-G and IV-H: a metric that looks decisive can be insensitive to exactly the effect it is being used to argue for.

B. PEDESTRIAN AND AGENT TRAJECTORY PREDICTION

A large part of the literature predicts pedestrians as points on the ground plane. Social LSTM [4] introduced neural social pooling, Social GAN [22] made the prediction explicitly multimodal, and Social-STGCNN [23] replaced recurrence with a spatio-temporal graph. Trajectron++ [5] incorporates heterogeneous agents and dynamic feasibility, and AgentFormer [24] models social and temporal dimensions jointly with attention. Goal-conditioned formulations predict an endpoint first and the path second [25], [26]. Diffusion has since become a standard generator in this space, with MID [6] treating prediction as progressive denoising of indeterminacy and LED [11] accelerating sampling through a learned initialiser. On the vehicle side, MTR [27] couples global intention localisation with local refinement, and UniTraj [12] studies

cross-dataset generalisation across nuScenes, Argoverse 2 and Waymo Open Motion. Rudenko *et al.* [28] and Karle *et al.* [29] survey the human and vehicle branches respectively.

All of these predict root position. Our root-trajectory evaluation in Section IV-G deliberately makes use of their metrics (average and final displacement error, reported as min-of-K) so that the conditioning question can be asked on the axis this literature has already validated. The difference is that our model must additionally produce body articulation, and, as Section IV-B shows, the two components of the output are not of equal quality.

C. HUMAN MOTION SYNTHESIS AND LATENT DIFFUSION

Generating 3D human motion from a high-level specification progressed from action labels [30] to free-form language [31], supported by the HumanML3D corpus and its 263-dimensional redundant motion representation [32], which we adopt. Denoising diffusion [33], with deterministic DDIM sampling [34] and classifier-free guidance [35], now dominates the area: MDM [8] and MotionDiffuse [9] denoise raw motion, while MLD [7] moves the process into the latent space of a motion VAE [36], following latent diffusion in images [15]. MLD is our backbone: we retain its VAE architecture (retrained on our data and frozen thereafter), transformer denoiser [37] and sampler, and replace the text condition with vehicle ego odometry. Alternative generators include discrete token prediction [38], generative masked modelling [10], motion-as-language formulations [39], retrieval augmentation [40], and the reuse of a pretrained motion diffusion model as a generative prior for downstream tasks [41].

Closest in spirit to our conditioning problem is spatially controlled motion generation. GMD [13] guides generation with ground-plane trajectory constraints, and OmniControl [14] controls any joint at any time through a combination of spatial guidance and a learned control branch. These methods constrain the generated human directly: the control signal specifies where that person's own body should be. Our condition differs in kind. The ego trajectory belongs to a *different* agent, and never determines the output since it only shapes a distribution over plausible reactions. This distinction is what makes conditioning so difficult to verify here, and is the reason a distributional score alone cannot establish it.

D. INTERACTION- AND SCENE-CONDITIONED MOTION

Motion conditioned on external context is usually conditioned on static geometry. PROX [42] resolves pose ambiguity using scene constraints, SAMP [43] generates stochastic scene-aware movement and interaction, HUMANISE [44] pairs language with 3D scenes, and TRUMANS [45] scales dynamic human–scene interaction data and modelling. For multi-agent settings, InterGen [46] generates two interacting people with mutual attention. Our setting sits between these: the conditioning agent is dynamic rather than static geometry, but unlike two-person interaction it is observed and fixed, which permits a simpler asymmetric design in which the ego trajectory enters the denoiser as a fixed token sequence.

E. FROZEN ENCODERS IN CONDITIONAL GENERATIVE MODELS

A recurring pattern in conditional image generation is that frozen conditioning encoders preserve generative diversity better than jointly fine-tuned ones. Stable Diffusion [15] keeps its text encoder frozen and IP-Adapter [16] and ControlNet [17] attach conditioning adapters to a frozen generative backbone. Conversely, methods that fine-tune the generative stack for condition fidelity, such as DreamBooth [47], are known to trade away diversity and require explicit prior-preservation regularization to recover it. Our freeze-versus-unfreeze ablation (Section V-D) supplies direct evidence for the same phenomenon in motion generation, and further shows that the degradation is not uniform across metrics: unfreezing improves mean trajectory error while collapsing per-condition diversity.

F. PSEUDO-GROUND-TRUTH MOTION LABELS

Large-scale driving datasets [18], [19], [48] provide vehicle odometry and 3D pedestrian boxes, but usually no body pose. JRDB [49] adds human-centric annotation in a robotics setting, and OmniRe [50] reconstructs dynamic urban scenes including human actors. Body labels for driving data must therefore be estimated, typically by fitting the SMPL body model [51] to images [52] and video [53]–[55], with world-grounded variants disentangling camera and human motion [56], [57]. Mocap corpora such as AMASS [58] supply the real-motion statistics against which such estimates are usually compared.

Estimated motion is known to be physically imperfect, and foot sliding is the canonical symptom. Rempe *et al.* [59] recover contact and dynamics from monocular video, Xie *et al.* [60] and Zhang *et al.* [61] impose physical and learned motion priors, PhysDiff [62] projects diffusion samples through a physics simulator, and UnderPressure [63] detects contact and cleans footskate from pressure-derived supervision. This literature treats sliding as a property of a *model* to be corrected at generation time. Section IV-B reports the less comfortable case in which the sliding is a property of the *labels*, so that no generator trained on them can be penalized for reproducing it, and Section IV-B accordingly locates the fix in relabeling rather than in an added loss.

G. EVALUATING CONDITIONAL GENERATIVE MODELS

Fréchet-distance scores compare a set of generated samples to a set of real ones through the statistics of a pretrained feature extractor [64], and the motion literature adopts this wholesale, substituting a motion encoder for the image one. The known failure modes of such scores are well cataloged: sensitivity to sample size and estimator bias [65], to image resizing and preprocessing [66], and to the feature extractor's own class structure [67]. Alternatives replace the Gaussian assumption

with kernel distances [68], richer embeddings [69], or explicit precision and recall [70]. Related critiques target the Inception Score [71] and its video analogue [72].

Every one of these critiques concerns how faithfully the score measures the *marginal* distribution. The problem we document is orthogonal and, to our knowledge, has not been made explicit for conditional motion generation: a score can measure the marginal perfectly and still be silent about whether the condition was used at all. We show this is not a hypothetical since on our benchmark a separately trained unconditional model matches the conditional one, and under interaction-aware sampling beats every conditional model in the paper, while remaining at chance on every conditioning measure (Sections IV-F and IV-G).

III. METHOD

A. PROBLEM FORMULATION

Given a vehicle ego trajectory $\mathbf{e} \in \mathbb{R}^{T\times 2}$ (2D position over $T = 196$ timesteps), we aim to generate a plausible 3D pedestrian body motion $\mathbf{x} \in \mathbb{R}^{T\times 263}$ in the HumanML3D representation [32]. The 263D representation encodes root velocity, root angular velocity, joint positions, velocities, and foot contact labels, fully parameterizing a SMPL-like body in motion.

We model the conditional distribution $p(\mathbf{x} \mid \mathbf{e})$ using a latent diffusion model that denoises within a pre-trained VAE's latent space.

B. MOTION VAE

We use the MLD VAE [7], a Transformer encoder-decoder that compresses $T \times 263$ motion to a latent $\mathbf{z} \in \mathbb{R}^{L\times 256}$ with $L = 4$ latent tokens (L is a hyperparameter, we find $L = 4$ significantly outperforms $L = 8$, see Section V). Rather than reusing the released HumanML3D checkpoint, we train the VAE from scratch on the pedestrian motions of our training split (AVA, nuScenes and Waymo, see Sec. IV) for 6,000 epochs, with motion features normalized by the mean and standard deviation of these three sources. HumanML3D itself is not part of its training data. The VAE is then held fixed throughout ego-encoder pretraining and diffusion training.

C. EGO ENCODER

The ego trajectory $\mathbf{e} \in \mathbb{R}^{196\times 2}$ is encoded by a lightweight transformer encoder:

$$\mathbf{c} = \mathrm{EgoEncoder}(\mathbf{e}) \in \mathbb{R}^{196\times 256}. \tag{1}$$

The encoder consists of a linear projection from 2D to 256D, followed by sinusoidal positional encodings, and $N = 4$ transformer encoder layers with 4 heads. Unlike the pooled variant (Pooled), which applies mean-pooling to produce a single token $\mathbf{c} \in \mathbb{R}^{1\times 256}$ similar to the language conditioning variant, our encoder preserves the full temporal sequence.

a: Ego encoder pretraining.

Before diffusion training, the ego encoder is pretrained with a regression-based alignment objective that distills the paired motion's VAE latent into the ego encoding. Specifically, we align the mean-pooled ego encoding $\bar{\mathbf{c}} = \mathrm{MeanPool}(\mathbf{c}) \in \mathbb{R}^{256}$ with the mean-pooled VAE posterior mean $\bar{\mathbf{z}} \in \mathbb{R}^{256}$ (computed from the paired motion) using a combined MSE and cosine objective:

$$\mathcal{L}_{\mathrm{pretrain}} = \|\bar{\mathbf{c}} - \bar{\mathbf{z}}\|^2 + \lambda_{\cos}\big(1 - \cos(\bar{\mathbf{c}}, \bar{\mathbf{z}})\big), \tag{2}$$

where the cosine term encourages directional alignment in addition to magnitude. After pretraining, the encoder achieves validation MSE ≈ 1.92, matching the pooled variant (which uses an additional 2-layer projection head without improvement), confirming that the projection head adds no alignment capacity.

b: Frozen encoder.

During diffusion training, the ego encoder parameters are *frozen*. This preserves the encoder's learned alignment while ensuring the generated motion distribution is not over-constrained to particular ego conditions. We ablate this choice in Section V.

The overall architecture is shown in Figure 1.

D. DIFFUSION DENOISER

We use the MLD denoiser as a self-attention transformer that operates over a single concatenated token sequence. At each denoising step, the $L = 4$ noisy latent tokens $\mathbf{z}_t$, the timestep embedding $\mathbf{t}_{\mathrm{emb}}$, and the full $T = 196$-token ego conditioning sequence $\mathbf{c}$ are concatenated into one sequence of length $L+1+T = 201$ and processed by the transformer; the first L output tokens are read out as the predicted noise:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c}) = \mathrm{Transformer}\big([\mathbf{z}_t,\, \mathbf{t}_{\mathrm{emb}},\, \mathbf{c}]\big)_{1:L}, \tag{3}$$

where the transformer applies self-attention over the joint sequence and the sampler computes $\mathbf{z}_{t-1}$ from $\hat{\boldsymbol{\epsilon}}$. Because every ego token participates in the same self-attention as the latent tokens, each denoising step has access to the full ego trajectory at per-timestep resolution: the denoiser can attend to specific moments in the ego trajectory (e.g., a braking event or sharp turn) rather than a global average. This is the key distinction from the pooled baseline (Pooled), whose denoiser sees only a single mean-pooled ego token ($L + 2$ tokens in total) and therefore has no temporal resolution over the ego trajectory.

The added cost over the pooled baseline is modest at this scale: self-attention over 201 tokens versus $L + 2$. A natural alternative is a *cross-attention* denoiser, in which the L latent tokens act as queries and the ego sequence as keys/values, reducing the attention cost to $\mathcal{O}(L \times T)$ rather than $\mathcal{O}((L + T)^2)$. We evaluate this variant in Section V and find it performs *comparably*, indicating that the benefit comes from exposing the denoiser to the full ego sequence, not from the specific attention mechanism used to inject it.

E. CLASSIFIER-FREE GUIDANCE

We train with classifier-free guidance [35]: for 10% of training samples, the input ego trajectory is zeroed before encoding,

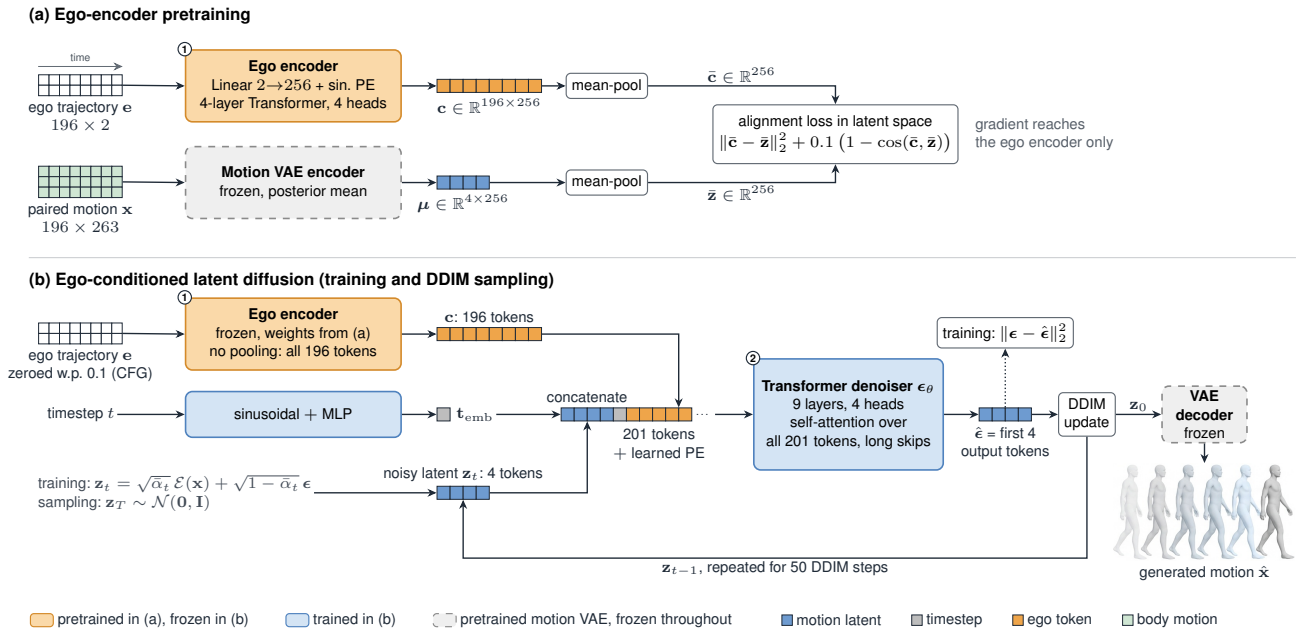


FIGURE 1: EgoPed architecture. **(a) Ego-encoder pretraining.** The ego encoder (1) is trained so that its mean-pooled output $\bar{\mathbf{c}}$ matches the mean-pooled posterior mean $\bar{\mathbf{z}}$ of the frozen motion VAE for the paired motion. **(b) Latent diffusion.** The ego encoder is *frozen* and keeps all 196 per-timestep tokens; for classifier-free guidance the ego input is zeroed for 10% of training samples. At every denoising step, the $L = 4$ noisy latent tokens $\mathbf{z}_t$, the timestep embedding and the 196 ego tokens are concatenated into one 201-token sequence and processed by a self-attention transformer denoiser (2), whose first four output tokens predict the noise. Sampling runs 50 DDIM steps and decodes $\mathbf{z}_0$ with the frozen VAE decoder; only the denoiser is trained in (b). Section V-E ablates an alternative cross-attention denoiser.

so the encoder's response to the all-zero trajectory serves as the unconditional embedding (the same construction is used for the unconditional branch at inference). At inference, the guided score is:

$$\hat{\boldsymbol{\epsilon}}_t = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \emptyset) + w \cdot \big(\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \mathbf{c}) - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \emptyset)\big), \tag{4}$$

with guidance scale w. We find $w = 10$ optimal for EgoPed (Section IV).

F. TRAINING OBJECTIVE

The denoiser is trained with the standard DDPM noise-prediction objective:

$$\mathcal{L} = \mathbb{E}_{\mathbf{z}_0, \boldsymbol{\epsilon}, t}\big[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c})\|^2\big]. \tag{5}$$

We use DDIM [34] for fast inference with 50 denoising steps.

IV. EXPERIMENTS

A. DATASET

As there are no datasets available for the problem investigated in this work, we generated a dataset for ego-conditioned motion generation from two complementary sources: existing large-scale autonomous driving datasets (nuScenes [18], Waymo [19]), which supply scale and behavioral diversity, and a small, deliberately *staged* set of high-interaction scenes recorded with our sensor-equipped research vehicle AVA, which targets the close-range vehicle–pedestrian encounters that are rare in the public datasets. The vehicle is equipped with seven 128-layer LiDARs and eight RGB cameras for full 360° sensor coverage around the vehicle. To obtain scenes with meaningful interaction between the vehicle and a pedestrian, we staged 17 highly interactive scenarios which we mirror yielding 34 scenarios in total. The recorded data is brought into nuScenes format and objects in the scene are annotated with 3D bounding boxes and orientation labels using MS3D [73], [74]. Fig. 2 shows the vehicle, a Mercedes-Benz S-Class built up by the research group. Most sensors sit on a roof-mounted aluminum rack: LiDARs (Ouster OS0, OS1, and OS2) at the rack corners and on a raised center mast, RGB camera clusters facing forward, to the sides, and the back (Basler ACE2), and a GNSS receiver with a dual antenna setup (Novatel OEM7) for localization. Further LiDARs are mounted at the front and back bumper. The ego odometry used as conditioning signal is taken from the vehicle's localization. The interaction scenarios were recorded in Munich by the authors who were asked to perform interactive scenes, e.g. crossing in front of the vehicle, stopping at the kerb, hesitating, turning around while the vehicle approaches, breaked, or turned.

With the data in nuScenes format and the single pedestrian annotated with 3D bounding boxes, we extract 3D human poses in SMPL [51] format using an adapted version of OmniRe [50] from all three data sources: AVA, nuScenes,



FIGURE 2: Our research vehicle AVA, used to record the staged high-interaction scenes. **Left:** top view of the roof-mounted sensor rack. **Right:** side view. Photos: Copyright by Johanna Weber.

and Waymo Open Dataset. We estimate poses for every pedestrian instance from all available camera images, using the projected 3D bounding box to localize the person in each view. Per-instance pose estimates from different cameras are then stitched into a single continuous sequence using the extrinsic calibration of the camera rigs. Finally, the SMPL pose sequences are converted to the 263D HumanML3D motion representation [32], which retargets every sequence onto a single canonical skeleton. Body shape is therefore not represented in our data (Sec. IV-B). In addition to the pedestrian pose sequences, we extract the synchronized ego-vehicle trajectory from each scene's odometry. A single automated pipeline processes every (scene, pedestrian) pair end-to-end: it loads the annotated nuScenes-format scene, extracts and stitches the per-instance SMPL body sequence, resamples both the body motion and the ego odometry to a common 20 fps grid, and expresses both streams in a *pedestrian-centric* coordinate frame (translation- and yaw-normalized to the pedestrian) so that conditioning is invariant to the absolute world pose of the scene. Each resulting sample is stored as a self-contained record containing the pedestrian's 22 body-joint positions (`ped_in_ped_frame` $\in \mathbb{R}^{T\times 22\times 3}$), the ego trajectory in the same frame (`ego_in_ped_frame` $\in \mathbb{R}^{T\times 3}$), and the derived 263D HumanML3D motion vector (`vectors_263` $\in \mathbb{R}^{T\times 263}$), where the native sequence length T varies per scene and is windowed to 196 frames at training time (see the interaction-aware cropping procedure below).

Using all three data sources we obtain **11,954** paired ego–pedestrian motion samples in total, split by scene into 9,540 training and 2,414 validation samples with no scene overlap between the two splits. Per-source counts are given in Table 1. Waymo and nuScenes dominate the sample count, while the AVA recordings contribute a small but deliberately high-interaction subset of staged vehicle–pedestrian encounters.

Each sample is a paired sequence of ego odometry $\mathbf{e} \in \mathbb{R}^{196\times 2}$ and pedestrian body motion $\mathbf{x} \in \mathbb{R}^{196\times 263}$, both expressed in the pedestrian-centric frame.

TABLE 1: Dataset composition. Each sample is a paired sequence of ego odometry (196×2) and pedestrian body motion (196×263), extracted from AVA, nuScenes, and Waymo. Train/validation splits are disjoint at the scene level; the validation split is the evaluation set used for all reported metrics.

Source	Train	Validation	Total
AVA (ours)	27	7	34
nuScenes	3,637	943	4,580
Waymo Open Dataset	5,876	1,464	7,340
Total	9,540	2,414	11,954

a: Interaction scoring.

Rather than hard-filtering samples, we assign each scenario a scalar *interaction score* that quantifies how strongly the pedestrian and ego vehicle interact, and use it both to bias training-time sampling and to guide cropping. Let $\mathbf{r}_t, \mathbf{g}_t \in \mathbb{R}^2$ denote the ground-plane (XZ) positions of the pedestrian pelvis and the ego vehicle at frame t. The score combines three factors:

$$s = s_{\text{travel}} \cdot s_{\text{prox}} \cdot s_{\text{bearing}}, \tag{6}$$

with the three factors

$$s_{\text{travel}} = \sum_t \|\mathbf{r}_{t+1} - \mathbf{r}_t\|_2, \tag{7}$$

$$s_{\text{prox}} = \frac{1}{T}\sum_t \mathbb{1}[\|\mathbf{g}_t - \mathbf{r}_t\|_2 < 5\,\text{m}], \tag{8}$$

$$s_{\text{bearing}} = 1 + \frac{1}{180^\circ}\sum_t |\Delta\phi_t|, \tag{9}$$

measuring pedestrian travel, the fraction of frames spent in proximity to the ego vehicle, and the swept relative bearing. Here $\mathbb{1}[\cdot]$ denotes the indicator function, $\phi_t = \text{atan2}(\mathbf{r}_t - \mathbf{g}_t)$ is the pedestrian's bearing relative to the ego vehicle, and $\Delta\phi_t$ is the (wrap-corrected) inter-frame change in bearing. Intuitively, a scenario scores highly when the pedestrian moves substantially, spends many frames within 5 m of the vehicle, and sweeps a large relative bearing (e.g., crossing in front of the vehicle).

b: Interaction-weighted sampling.

The pipeline supports drawing training samples with a weighted random sampler whose weights are the interaction scores s (floored at 0.01 so low-interaction scenarios are still occasionally seen, then normalized to sum to the dataset size), concentrating gradient updates on the ego-relevant portion of the data distribution without discarding any samples. The VAE, the pooled baseline (Pooled), and our best system (EgoPed-IA) train with this weighting; the EgoPed reference model trains with uniform sampling, and we disentangle the two effects with a full 2×2 ablation in Section V-B. Evaluation always uses uniform (unweighted) sampling.

c: Interaction-aware cropping.

To bound the generation problem to a fixed horizon, we crop each scenario to $T = 196$ timesteps (≈40% of sequences are longer). The pipeline supports centering the crop on the *closest-approach frame* $t^\star = \arg\min_t \|\mathbf{g}_t - \mathbf{r}_t\|_2$, with uniform jitter of $\pm 25\%$ of the window length so the moment of closest approach is not memorized at a fixed position (the window is clamped to lie within the sequence). The VAE, Pooled, and EgoPed-IA use interaction-centered crops; the EgoPed reference model uses random windows (ablated in Section V-B). In both modes, ego and motion streams are cropped with a shared index so they stay temporally aligned, and scenarios shorter than 196 frames are zero-padded at the end with their true length recorded for masking.

B. PSEUDO-GROUND-TRUTH POSE QUALITY

Our pose labels are OmniRe estimates rather than measured 3D ground truth, and no reference source (mocap or manual annotation) exists for these scenes. We therefore audit what can be checked without a reference by relying on internal physical consistency over all 14,172 extracted sequence files (13,958 of them long enough to score). The audit script is released with the code. This population is deliberately the full extraction, a superset of the 11,954 samples the benchmark splits retain (Table 1), so the audit characterizes the label pipeline rather than only the sequences we train on. Table 2 reports the result, and it materially qualifies how our metrics should be read.

Three things stand out. First, **body shape was never estimated**: every sequence in all three sources shares one skeleton to within 10^{-5} relative bone length, so the corpus contains a single canonical body rather than per-person shapes. This also means bone-length and left/right-symmetry checks carry no quality signal here, since rigidity is imposed by the representation rather than recovered from images. Second, **the feet do not plant**. During locomotion at least one foot should be stationary in world coordinates; taking the slowest of the four foot joints per frame and dividing by root speed gives a median ratio of 0.97, where 0 denotes clean footfalls and 1 a body sliding as a rigid unit, and 94.6% of moving sequences never hold a foot still for more than 5% of their frames. This check is internal and has no domain confound. Third, **articulation is damped** relative to real mocap: against HumanML3D's own per-dimension statistics, root translation is *more* variable than mocap ($1.29\times$) while root-relative joint positions, root height and turning are far less ($0.44\times$, $0.26\times$, $0.37\times$). Part of that gap is content rather than quality as HumanML3D spans every kind of motion and ours is people walking near roads, so we read it as corroborating the foot-skate result rather than as an independent measurement.

a: Where the sliding lives.

Because the artifact is in the labels, it is worth asking whether it is also in the model, and whether a training-side fix could remove it. We apply the identical foot-skate measure at three points of the pipeline on the same held-out conditions (Table 3). The labels slide at 0.96. A VAE round-trip of those labels, which involves no diffusion model at all, already slides at 0.94, and samples from the trained models slide at 0.93–0.94. Our models neither amplify nor repair the artifact. They are faithful fits to a target that slides, which is the expected behavior of a well-trained generative model and not an independent failure. And because the frozen VAE reproduces the sliding on its own, the artifact is embedded in the latent space that every denoiser in this paper decodes through, so a contact or foot-skate penalty applied to the denoiser would be optimizing against its own decoder. Removing the sliding therefore requires correcting the labels and retraining the VAE, which changes the target distribution and invalidates every FID value reported here, so we do not attempt it for this paper.

We did run a feasibility pilot, because the more damaging possibility is that the latent bottleneck itself cannot represent planted feet. Anchoring footfalls to the measured root path (preserving the root and upper body, re-solving the legs by inverse kinematics) removes the sliding from the labels, and on 1,418 such sequences the original VAE reconstructs them at a skate ratio of 0.75 whereas a VAE fine-tuned for 400 epochs on the corrected labels reconstructs at 0.15. The representation is therefore not the obstacle, and relabeling is a viable if expensive direction. We note also that any correction that preserves the tracked root must re-synthesize the legs, replacing estimated poses with procedurally generated gait, which is a different trade rather than a strictly better ground truth.

The picture is a lightly articulated canonical body translating along a well-tracked trajectory. That asymmetry maps cleanly onto our conclusions. The evidence that conditioning is *used* (root ADE/FDE (Sec. IV-G) and the stop/walk probe (Sec. IV-H)) is computed from the root trajectory, the component this audit finds trustworthy, so those conclusions are unaffected. FID is the exposed quantity: it scores body pose against this same damped target, which makes the marginal easier to fit than the visual richness of real pedestrians would suggest and helps explain why an ego-blind prior can reach FID 1.43 (Table 4). Absolute FID values here should be read as within-benchmark comparisons only, never as a claim about pose realism.

C. EVALUATION PROTOCOL

We report four metrics, following the HumanML3D evaluation protocol [32]:

- **FID** (Fréchet Inception Distance): computes the feature-space distance between the generated and real motion distributions, using the pretrained move encoder (stride-2 Conv1D) and motion encoder (BiGRU) from [32]. Lower is better.
- **R-Precision@1**: top-1 retrieval accuracy of the ego condition from batches of 32 candidates (1 positive + 31 negatives), using cosine similarity between the mean-

TABLE 2: Reference-free audit of the pseudo-ground-truth poses (13,958 sequences, 10,900 contain locomotion). Upper block: internal physical consistency, no external reference and no domain confound. Lower block: ratio of our per-dimension standard deviation to HumanML3D's real-mocap statistics, by block of the 263-D representation; values below 1 indicate less variability than mocap, though part of that gap reflects the narrower motion content of driving scenes.

Internal consistency check	Result
Distinct skeletons among all sequences	1
Max. relative bone-length deviation	6.3×10^{-6}
Foot-skate ratio, median (0 = plant, 1 = slide)	0.97
AVA / nuScenes / Waymo	0.73 / 0.99 / 0.95
Never plant a foot (>5% of moving frames)	94.6%
Near-static legs (ankle swing $<$ 5 cm)	14.1%
Ground penetration $>$ 5 cm	1.2%
Variability vs. mocap (ours / HumanML3D std)	**Ratio**
Root linear velocity	1.29
Joint rotations	0.68
Root-relative joint positions	0.44
Root rotational velocity	0.37
Root height	0.26

TABLE 3: Foot-skate ratio measured at three points of the pipeline (held-out split, same metric as Table 2; 0 = clean footfalls, 1 = the body sliding as a rigid unit). The VAE round-trip involves no diffusion model, so the sliding is already present in the latent representation; the generative models inherit it without amplifying it.

Pipeline stage	Skate	Planted (%)
Pseudo-ground-truth labels	0.96	0.6
VAE round-trip (encode–decode)	0.94	0.0
Generated, EgoPed	0.94	0.0
Generated, EgoPed-IA	0.93	0.0
Generated, unconditional prior	0.84	0.6

pooled VAE latent (from generated motion) and the ego encoder output. Higher is better.

- **MultiModality (MM)**: average pairwise L2 distance between 10 generated samples for the same ego condition, measuring diversity per condition. Higher indicates more diverse generation per condition, but is only meaningful at comparable conditioning fidelity since an unconditional model maximizes MM trivially (cf. Table 4), so MM should be read jointly with R-Precision.
- **Diversity**: average pairwise L2 distance in feature space across all generated samples, measuring population-level diversity. Values *closer to the ground-truth diversity* (5.496 on our evaluation set) are better.

All reported values are the mean over **3 independent replications** with 95% confidence intervals. Unless stated otherwise, metrics are computed on the full validation split (2,414 samples), which is also used for checkpoint selection, under a *unified evaluation pipeline*: identical eval-time data processing (uniform sampling, random 196-frame windows) for every system, regardless of its training-time data pipeline. We distinguish two kinds of metric values. *Training-time* metrics are logged by a single-pass validation run every 100 epochs during training and are used only for checkpoint selection and training curves. All reported results are *definitive* 3-replication evaluations of the selected checkpoint under the unified pipeline. To control for selection bias, we additionally construct a *held-out test split*: the validation scenes are partitioned into two scene-disjoint halves, one reserved purely for held-out reporting (1,190 samples from 521 scenes, never used for any model or checkpoint selection of our main system). Our best system reproduces its headline result on this held-out split (FID $= 3.34 \pm 0.19$ vs. 3.39 ± 0.18 on the full validation split), confirming the result is not an artifact of evaluating on the selection set. The conditioning-mechanism ablation (Section V-E) is reported on this split. Note that FID and Diversity use a fixed, pretrained evaluator shared by all systems, so they are directly comparable across architectures. R-Precision, in contrast, retrieves with each model's *own* ego encoder embedding. Cross-architecture R@1 comparisons (e.g., pooled vs. full-sequence encoders) should therefore be read as indicative rather than exact, as the retrieval space itself differs.

D. IMPLEMENTATION DETAILS

a: Ego encoder.

4-layer transformer encoder, 4 attention heads, 256D hidden dimension (1024D feed-forward), sinusoidal positional encoding for $T = 196$ timesteps. Input: 2D ego position sequence, projected linearly to 256D. No projection head (we verified the 2-layer MLP projection head does not improve val MSE: 1.919 vs. 1.920). Pretraining: AdamW (lr $= 1 \times 10^{-4}$, weight decay 10^{-4}), 200 epochs, batch size 64, linear warmup followed by cosine annealing, gradient clipping at max-norm 1.0.

b: Diffusion model.

MLD denoiser in the self-attention configuration (attending over the concatenated latent, timestep, and ego tokens): 9 layers, 4 attention heads, 256D token dimension, 1024D feed-forward, latent sequence length $L = 4$. The cross-attention variant of Section V-E uses the same hyperparameters in a cross-attention configuration (latent tokens as queries, ego sequence as keys/values). Diffusion: 1000 DDPM training steps (scaled-linear β schedule), DDIM inference (50 steps, $\eta = 0$). CFG dropout rate: 10%. Optimizer: AdamW, lr $= 1 \times 10^{-4}$ (constant), 5000 total epochs. Batch size: 128. Training hardware: 4 NVIDIA H100 (80 GB) on an academic

HPC cluster with training runs in segments of up to 24h wall time.

c: Checkpoint selection.

We select checkpoints based on validation FID (not training loss). For EgoPed, epoch 3399 is best. We confirm this with the full training trajectory and definitive 3-replication evaluation. For Pooled, epoch 4399 is best despite continued training to epoch 5000.

d: Evaluation.

All evaluations use 3 independent generation passes, 300 samples for Diversity, and 100 conditions $\times$ 10 repeats for MultiModality. Each replication regenerates the full test set with fresh sampling noise. The diffusion sampler is stochastic across replications while the data pipeline and evaluator are fixed (training uses seed 1234). Confidence intervals are reported as $\pm$ CI (half-width of the 95% interval across replications).

E. BASELINES

- **Pooled (our baseline)**: pooled ego encoder (transformer + mean-pool $\rightarrow$ single 256D token), self-attention denoiser (concatenated self-attention over $[\mathbf{z}, \mathbf{c}_{\text{pool}}, \mathbf{t}]$). Best at epoch 4399, CFG = 5.
- **Latent-8**: Same as Pooled but with latent dimension $L = 8$ (larger VAE). Best at epoch 4399, CFG = 5.

We additionally report four non-learned or reference baselines (Table 4) that bound the metric space. **Unconditional MLD** runs our EgoPed model (epoch 3399) with every ego trajectory zeroed. Because CFG training dropout uses the same zeroed-trajectory construction, both guidance branches receive identical embeddings and guidance cancels exactly, so guidance cancels and the model falls back on whatever it learned to produce without the ego. We report it because the literature uses this construction, not because it is a well-trained prior. A *separately trained* unconditional model (identical recipe, ego dropped at every training step, 5,000 epochs, evaluated through the same ego-zeroing at guidance scale 1) reaches FID 3.24 $\pm$ 0.13 which is statistically indistinguishable from the full-sequence model (3.39 $\pm$ 0.18). FID therefore does not by itself credit ego conditioning: the ego-zeroed branch of a conditional model is merely a weak unconditional generator, and a well-trained prior matches the conditional model's *marginal*. Repeating the control on the interaction-sampling data recipe is more striking still: an unconditional model trained that way reaches FID 1.43 $\pm$ 0.18 while being demonstrably ego-blind (ADE 3.00, behavioral separation 0.02, R@1 at chance). Interaction sampling is therefore largely a *marginal-quality* intervention: it improves the prior (3.24 $\rightarrow$ 1.43) more than it improves the conditional model (3.39 $\rightarrow$ 2.88), so EgoPed-IA's FID advantage does not survive comparison against its own prior. Evidence that the conditioning is used comes from the evaluator-independent trajectory metrics (Sec. IV-G) and the behavioral probe (Sec. IV-H), which a model without ego access cannot match. **Retrieval** returns, for each test ego trajectory, the training motion whose ego trajectory is the nearest neighbour (32-point resampled L2). **VAE interpolation** decodes a random latent interpolation of two training motions. **Trajectory+kinematic** replaces all body features with the training mean, keeping either the ground-truth root velocity (*oracle*) or nothing (*mean*), simulating a trajectory-prediction-then-fitting pipeline with no learned body model. We further add two *learned* baselines trained on the identical data recipe with the same frozen ego encoder. **Ego$\rightarrow$motion regressor**: a deterministic sequence-to-sequence transformer regressing the 263-D features with masked MSE. **Raw-space diffusion (MDM-style)**: our diffusion model operated directly in the 263-D feature space without the latent VAE (a same-capacity raw-space variant in the spirit of MDM [8], not a faithful reproduction).Iits best checkpoint plateaus early and never approaches latent-space quality, justifying the latent backbone. **Pretrained text-to-motion MLD**: the released HumanML3D MLD checkpoint [7] generating from a text prompt instead of ego trajectory, evaluated on our held-out conditions. Because it is trained on a different corpus (mocap-quality everyday motion) its gap to our in-domain models conflates domain shift with the conditioning question. We therefore read it as evidence that off-the-shelf text-to-motion does not transfer to driving-scene pedestrians, rather than as a statement about text versus ego conditioning per se.

a: Information ladder: how much conditioning information is needed?

To separate *that* a condition helps from *how much* information it must carry, we train the same architecture with conditioning signals of increasing information content (Table 5) and evaluate all of them with the unified pipeline. Captions are synthesized deterministically from kinematics for all 14,172 sequences, and the information content is the entropy of the resulting template distribution. The *vehicle-only* vocabulary describes only what the ego-trajectory model sees (approach/pass/wait, side, proximity, ego speed and turning, timing of closest approach). The *oracle* vocabulary additionally verbalizes the pedestrian's own ground-truth behavior (gait, turning, stopping). It is therefore an upper bound, not a competitor. Text models use 10% caption dropout and CLIP text features, and are evaluated at the same CFG scale as the trajectory models (a sweep over {1, 2.5, 5, 10} is monotone for the oracle model: 6.43, 5.71, 4.14, 2.82). The vehicle-only model is evaluated on a 3-epoch$\times$4-CFG grid (epochs 99, 3399, 4999; FID ranges from 5.12 at epoch 99 with CFG 5 to 6.67). Its epoch-matched checkpoint reaches FID 5.91, ADE 2.52 and behavioral separation 0.21 which constitutes roughly two thirds of the trajectory model's per-condition gains over the unconditional prior (ADE 2.99 $\rightarrow$ 2.32, separation 0.04 $\rightarrow$ 0.33) at a much worse marginal, so a coarse verbal description of the vehicle carries real but incomplete ego information, and the

continuous trajectory supplies the rest on both axes. Its FID-best checkpoint (epoch 99, FID 5.12) is behaviorally ego-blind (separation 0.05), reproducing within a single model the dissociation between FID and conditioning.

F. MAIN RESULTS

Table 6 shows the definitive evaluation results. EgoPed achieves FID $= 3.392 \pm 0.179$ at epoch 3399 and CFG $= 10$, a **40% improvement** over the pooled baseline under the unified evaluation pipeline (FID $= 5.620 \pm 0.117$ at CFG $= 5$), and EgoPed-IA improves this further to 2.880 ± 0.084. The improvement is significant: the confidence intervals are non-overlapping (EgoPed CI: $[3.213, 3.571]$; Pooled CI: $[5.503, 5.737]$).

Interestingly, R-Precision@1 for EgoPed (0.548) is *lower* than Pooled's baseline (0.665). Three factors make cross-model R@1 comparisons unreliable: (1) EgoPed generates more diverse motions per ego condition (MM: 3.956 vs. 3.708), spreading the generated distribution in feature space and naturally reducing retrieval accuracy. (2) the R@1 metric measures alignment between generated VAE latents and ego encoder outputs. As the denoiser learns richer dynamics, this alignment metric underestimates the true conditioning fidelity. (3) R@1 retrieves with each model's *own* ego encoder (Section IV-C), so the pooled and full-sequence encoders define retrieval tasks of different intrinsic difficulty. The next section settles the question with an evaluator-independent metric: on actual trajectory-following, EgoPed conditions *better* than Pooled, not worse.

G. EVALUATOR-INDEPENDENT CONDITIONING: TRAJECTORY ADE/FDE

To measure conditioning fidelity without any learned evaluator or model-specific embedding, we compute root-trajectory displacement errors: for each held-out test condition we generate $K{=}5$ motions, recover the pelvis ground-plane trajectory, and compare with the ground truth over the true sequence length (ADE/FDE in meters, min-of-K is standard for stochastic predictors).

Three controls sharpen this reading. A *separately trained* unconditional model, whose FID (3.24) is indistinguishable from EgoPed's, is 28% worse in ADE and 25% worse in FDE: identical marginals, no per-condition fidelity. An *oracle* text model that is told the pedestrian's own behaviour (Table 5) reaches ADE 1.88, approaching the L_2-optimal regressor while remaining generative. This bounds what a perfect description of the outcome would gain. Finally, EgoPed-IA, which is the best model by FID, is *worse* than EgoPed on every per-condition metric (ADE 2.66 vs. 2.36, minADE_5 1.73 vs. 1.31). Interaction-aware sampling improves the marginal but not the conditioning. Its FID-selected checkpoint is trained for a third as long (epoch 1299 vs. 3399). An epoch-matched checkpoint recovers part of the gap (ADE 2.58, minADE_5 1.53) but still trails EgoPed on every metric, so the effect is attributable to the sampling itself rather than to under-training.

The unconditional reference is worst on mean ADE/FDE, confirming that ego conditioning demonstrably improves trajectory-following (3.01 $\rightarrow$ 2.36 mean ADE). This shows that its deceptively low minADE_5 reflects raw sample diversity, not conditioning.

Figure 3 shows generated samples for three held-out ego conditions. For visualisation, the generated HumanML3D joint sequences are retargeted onto CARLA pedestrian skeletons and replayed in a CARLA town together with an ego vehicle that follows the conditioning trajectory [Mohan: Please describe the retargeting and scene-placement procedure, CARLA version and town briefly].

a: Significance.

Every model is scored on an identical 196-frame window per condition: the evaluation crop is derived deterministically from the sample identity rather than drawn from the global random stream, which otherwise makes the window depend on how much randomness a particular configuration happens to consume first. Marginal intervals in Table 7 are percentile bootstraps and several of them overlap, but because the conditions are shared the appropriate test is a paired bootstrap on the per-condition difference. All comparisons that matter are significant at the 95% level: full-sequence beats pooled by $0.097\,\text{m}$ ($[0.017, 0.175]$, $p = 0.017$), and does so for all three training seeds ($p = 0.048$ and $p < 10^{-4}$ for the other two). Both trained priors are worse than EgoPed by $0.650\,\text{m}$ and $0.656\,\text{m}$ ($p < 10^{-4}$), which is the conditioning effect and is roughly seven times the pooled-vs-full-sequence gap. EgoPed-IA is worse than EgoPed by $0.299\,\text{m}$ at its own best checkpoint and $0.274\,\text{m}$ at its second seed ($p < 10^{-4}$), so its per-condition deficit is not a sampling artifact. Vehicle-only text is worse than EgoPed by $0.123\,\text{m}$ ($p = 0.019$) while the oracle beats EgoPed by $0.478\,\text{m}$ ($p < 10^{-4}$). The one comparison that cuts against our recommendation is Unfrozen, which beats EgoPed by $0.098\,\text{m}$ ($[0.019, 0.178]$, $p = 0.015$). We return to it below, since Unfrozen buys that accuracy with a 34% collapse in per-condition diversity.

H. BEHAVIORAL VALIDITY: INTENTION DISTRIBUTIONS

Beyond kinematic fidelity, we probe whether generated motion couples to the ego context at the level of *behavioral decisions*. We derive an automatic stopping label from root trajectories (speed below $0.2\,\text{m/s}$ sustained for $\geq 1\,\text{s}$ which are heuristic thresholds, documented and spot-checked) and, for each held-out condition, compare the model's sampled stopping probability $\hat{p}(\text{stop} \mid \mathbf{e}) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{1}[\text{stop}(\mathbf{x}_k)]$, with $\mathbf{x}_k \sim p(\mathbf{x} \mid \mathbf{e})$ against the ground-truth label ($K{=}5$, the same samples as Section IV-G). We report the *separation* (mean $\hat{p}$ on ground-truth-stopping conditions minus mean $\hat{p}$ on walking conditions), the mean per-condition binary entropy of $\hat{p}$, and the Brier score.

TABLE 4: Reference baselines bounding the metric space (val split, same evaluator, normalization, and split lists as Table 6). All rows are mean ± 95% CI over 3 replications (the unconditional row resamples the diffusion model; the non-learned rows are reseeded end-to-end). Retrieval attains near-zero FID by copying real motions but is non-generative; trajectory+kinematic shows that following the root trajectory with a static body is catastrophic. The 5.18 row is *not* a prior: it is the under-trained null branch of a conditional model, and separately trained unconditional models reach 3.24 on the standard recipe and 1.43 with interaction sampling. FID differences across these rows therefore order marginal fit, not the use of the ego; see Sec. IV-G and Sec. IV-H.

Method	FID ↓	Diversity	R@1 ↑
Retrieval (NN ego)*	0.062 ±0.004	5.34	–
Trajectory+kinematic (mean)	49.60 ±0.087	0.46	–
Trajectory+kinematic (oracle)	47.67 ±0.085	0.72	–
Pretrained text-to-motion MLD [7]§	14.07	3.12	–‡
Ego→motion regressor (learned, deterministic)	35.47	3.47	–
Raw-space diffusion (learned, MDM-style)	26.43 ±0.098	3.55	–‡
VAE latent interpolation	7.91 ±0.033	5.82	–
Unconditional EgoPed (ego zeroed)	5.18 ±0.209	5.96	0.031†
Unconditional EgoPed (trained, p_{drop}=1)¶	3.24 ±0.130	5.91	0.031†
+ interaction sampling¶	1.43 ±0.183	5.76	0.031†
Pooled	5.620	5.82	0.665
EgoPed	3.392	5.79	0.548
EgoPed-IA (ours)	**2.880**	5.67	0.323

*Non-generative upper bound: copies real training motions, so low FID reflects memorization rather than conditioned synthesis. †Exactly chance level ($1/32 = 0.031$), confirming the R-Precision protocol: without ego information, retrieval is random. ‡R-Precision is undefined for these models (no shared embedding space with the ego encoder). ¶Trained with $p_{\text{drop}} = 1$ (never sees the ego) and evaluated through the same ego-zeroing path as the row above. Training-time validation feeds real ego to these models and is not a valid selector, so we evaluate three late checkpoints (epochs 3999/4499/4999) offline and report the best: epoch 4999 for the standard recipe (the three span 3.24–4.90) and epoch 4499 for the interaction-sampling recipe (span 1.43–2.67; even its worst checkpoint matches EgoPed-IA). §Official HumanML3D checkpoint driven by a fixed text prompt; we report the *best* of three prompts (a naive scene description, a HumanML3D-idiomatic phrase, and per-sample oracle captions derived from ground-truth motion), which spanned FID 14.07–24.84. Its output is converted from HumanML3D to our normalization before scoring with the shared evaluator.

EgoPed-IA couples behavior to the ego less than half as strongly as EgoPed (separation 0.15 vs. 0.33) and under-predicts stopping (0.14 vs. a base rate of 0.22). At a matched epoch the coupling improves (0.23) while the stopping calibration worsens further (0.13), consistent with the interaction-weighted training distribution shifting behavioral calibration rather than with under-training. Across Tables 7 and 8, EgoPed is the best-*conditioned* model.

Path-crossing labels (pedestrian/ego polyline intersection) occur at only a ∼2% base rate in our windows and are omitted from the primary analysis.

I. CFG SCALE ANALYSIS

We sweep CFG $\in \{5, 10, 15\}$ for EgoPed at epoch 3399 (Table 9). EgoPed's FID varies non-monotonically with guidance and has a clear optimum at CFG = 10 (3.842 → 3.392 → 3.547), whereas Pooled degrades monotonically as guidance rises (6.603 → 6.963 → 7.400). This indicates that full-sequence conditioning naturally provides a rich, overdetermined conditioning signal. The denoiser attends to 196 ego tokens, so higher guidance amplifies an already high-fidelity signal rather than forcing mode collapse onto a coarse pooled token.

V. ABLATION STUDY

A. LATENT DIMENSION: 4 VS. 8

To test whether a larger VAE latent space improves generation quality, we train a latent-8 VAE and ego-conditioned diffusion model (Latent-8). As shown in Table 6, Latent-8 achieves FID = 6.677 which is **19% worse** than Pooled's 5.620 at the same epoch and CFG (unified evaluation). R-Precision is essentially unchanged (Latent-8: 0.663 vs. Pooled: 0.665), confirming that the bottleneck for conditioning quality is the ego encoder, not the latent space.

The mechanism: the fixed-capacity denoiser must now model an 8D latent distribution instead of 4D, a strictly harder task, without any additional denoiser parameters. The marginal diversity improvement (Diversity: 5.887 vs. 5.819, +1.2%) does not compensate.

B. INTERACTION-AWARE SAMPLING: A TRAINING-DATA 2×2

The interaction-aware pipeline of Section 4.1 (score-weighted sampling + closest-approach cropping) is a training-*data*

TABLE 5: Information ladder (val split, unified pipeline, 3 replications; CFG 10 for all conditional models, CFG 1 for the unconditional one). FID orders the *marginals*, not the conditioning: a well-trained unconditional model matches the trajectory model, and an oracle that is told the pedestrian's behaviour matches EgoPed-IA. Whether a condition is actually *used* is measured in Secs. IV-G and IV-H. ADE (m) and behavioural separation (held-out split, $K{=}5$) show which signals are actually *used*: the trained prior is ego-blind despite matching EgoPed's FID, the oracle is told the answer, and interaction sampling improves the marginal but not the per-condition fidelity (bold: best non-oracle model). The vehicle-only row is the epoch-matched checkpoint; its FID-best checkpoint (5.12) does not use the condition. The dissociation is sharpest off the ladder: an unconditional model trained with EgoPed-IA's own interaction-sampling recipe reaches FID 1.43, better than every conditional model here, at ADE 3.00 and separation 0.02 (Table 4).

Conditioning signal	Information	FID ↓	ADE ↓	Separation ↑	Diversity
None (trained unconditional, $p_{\mathrm{drop}}{=}1$)	0 bits	3.24 ±0.130	3.01	0.05	5.91
Vehicle-only text (epoch-matched, 3399)	5.17 bits	5.91 ±0.055	2.48	0.21	5.67
Oracle text (+ ground-truth pedestrian behaviour)	6.09 bits	2.82 ±0.027	1.88	0.94	6.20
Ego trajectory (EgoPed)	continuous, 196×2	3.39 ±0.179	**2.36**	**0.33**	5.79
Ego trajectory + interaction sampling (EgoPed-IA)	continuous, 196×2	**2.88** ±0.084	2.66	0.16	5.67

TABLE 6: Main evaluation results under a *unified* evaluation pipeline (identical eval-time data processing for all systems; each method at its own optimal CFG, sweep in Table 9). All values are mean ± 95% CI over 3 replications. Best values in **bold**; for Diversity, values closer to the ground-truth reference are better. EgoPed-IA additionally uses interaction-aware training-data sampling (Section V-B); its FID replicates across two training seeds (2.880 and 2.671), whereas EgoPed's seed sensitivity is much larger — both are quantified in Section V-C.

Method	Epoch	CFG	FID ↓	R@1 ↑	MM ↑	Diversity
Pooled (baseline)	4399	5	5.620 ±0.117	**0.665** ±0.003	3.708	5.819
Latent-8	4399	5	6.677 ±0.135	0.663 ±0.002	3.374 ±0.356	5.887
EgoPed (ours)	3399	10	3.392 ±0.179	0.548 ±0.002	**3.956** ±0.406	5.794
EgoPed-IA (Ours, + interaction sampling)	1299	10	**2.880** ±0.084	0.323 ±0.004	3.371 ±0.361	5.671
Unfrozen (Ablation, unfrozen encoder)	3399	10	5.079 ±0.034	0.352 ±0.009	2.618 ±0.309	6.064
Ground truth (reference)	—	—	—	—	—	5.496

intervention, orthogonal to the conditioning architecture. We disentangle the two with a full $2{\times}2$: each conditioning architecture (pooled, full-sequence) trained with and without the interaction pipeline, all evaluated under the unified pipeline. Table 10 reports the four cells.

Two conclusions follow. First, the conditioning-granularity gap (≈2.7–2.8 FID in both columns) is roughly five times the pipeline effect, confirming that full-sequence conditioning is the primary driver. Second, the two interventions are approximately additive: interaction-aware sampling contributes a similar absolute gain to both architectures (-0.59 pooled, -0.51 full-sequence), and for the full-sequence model the best checkpoint additionally arrives at $1/3$ the epochs. The +pipeline full-sequence cell (EgoPed-IA) has the best marginal of the four while EgoPed remains the best-conditioned configuration (Sec. IV-G).

C. TRAINING-SEED SENSITIVITY

We retrain EgoPed from two additional seeds under its original recipe and select each seed's best checkpoint by training-time validation FID (the same protocol as the main results). Definitive best-checkpoint FID across the three seeds is 3.392, 4.719, and 4.803 (mean 4.30 ± 0.79): the headline seed is markedly better than the other two, so single-seed point estimates in this setting overstate precision, and we report the spread explicitly. The cross-*architecture* conclusions are unaffected since every EgoPed seed remains well below the pooled baseline (5.620), and the Pooled baseline itself proved training-unstable (two additional Pooled seeds were still above FID 10.9 when stopped at roughly 2,400 epochs, about half the epoch at which the reported seed peaked, at around four times lower throughput), meaning all our comparisons use the baseline's *best* seed and are conservative in the baseline's favor. Running the per-condition protocol on all three EgoPed seeds sharpens the point considerably. While FID ranges over 3.392–4.803 across those seeds, a relative spread of 33%, their trajectory error ranges only over 2.258–$2.375\,\mathrm{m}$ (5%) and their behavioral separation over 0.311–0.334 (7%). The per-condition instruments are several times more stable under reseeding than the metric the field

TABLE 7: Root-trajectory displacement errors (meters, held-out test split, K=5 samples per condition, N=1,190). EgoPed beats the pooled baseline on *all four* metrics (paired bootstrap over conditions: ADE, minADE, and minFDE significant) despite its lower R@1. This is direct evidence that the cross-model R@1 gap is an embedding-space artifact. The deterministic regressor minimizes expected L2 by construction yet is unrealistic (FID 35.5, Table 4). With only 5 samples EgoPed's minADE beats even this L2-optimal predictor. Mean ADE rewards mode concentration (collapsed Unfrozen scores best) while min-of-K rewards coverage — the two must be read together. Bold marks the best ego-trajectory-conditioned model per column. The unconditional and oracle rows are reference bounds (min-of-K rewards the diversity of an unconditional sampler and is not a conditioning test).

Model	ADE ↓	FDE ↓	minADE_5 ↓	minFDE_5 ↓
EgoPed	2.360 ±0.110	4.896 ±0.221	1.306	2.625
EgoPed, seed B	2.375 ±0.116	4.932 ±0.220	1.321	2.631
EgoPed, seed C	2.258 ±0.108	4.666 ±0.216	1.421	2.873
Pooled	2.456 ±0.114	5.032 ±0.230	1.493	3.013
Unfrozen	2.262 ±0.113	4.637 ±0.229	1.535	3.101
Regressor (deterministic, K=1)	1.800	3.671	—	—
EgoPed-IA (+ interaction sampling)	2.659 ±0.114	5.551 ±0.243	1.732	3.571
EgoPed-IA, epoch-matched to EgoPed (3399)	2.615 ±0.112	5.438 ±0.234	1.564	3.217
EgoPed-IA, seed 2 (best-val, epoch 899)	2.633 ±0.111	5.526 ±0.238	1.744	3.555
EgoPed-IA, seed 2 at epoch 1299	2.599 ±0.115	5.401 ±0.246	1.712	3.471
Unconditional (ego zeroed)	2.897 ±0.101	5.913 ±0.205	1.236	2.463
Unconditional (trained, p_{drop}=1)	3.010 ±0.100	6.136 ±0.199	1.233	2.445
+ interaction sampling (FID 1.43)	3.016 ±0.104	6.260 ±0.214	1.379	2.748
Vehicle-only text (epoch 3399)	2.483 ±0.108	5.033 ±0.209	1.349	2.695
Vehicle-only text, FID-best checkpoint (epoch 99)	2.703 ±0.095	5.517 ±0.197	1.310	2.556
Oracle text (told the pedestrian's behaviour)	1.882 ±0.133	3.822 ±0.266	1.285	2.552

reports, which is a further argument for reading conditioning off them. All three seeds beat the trained unconditional prior decisively and the pooled baseline significantly (paired bootstrap, $p \leq 0.048$), so the per-condition ranking does not depend on which seed we happened to report. EgoPed-IA, in contrast, *replicates*: a second seed trained with an identical recipe and selected by the same rule reaches FID 2.671 ± 0.031 (its best-validation checkpoint is epoch 899) against the reported seed's 2.880 ± 0.084, i.e. a seed mean of 2.78 ± 0.15 which is a far tighter spread than EgoPed's. The per-condition picture replicates as well: the second seed reaches ADE 2.63 and behavioral separation 0.13 (reported seed: 2.66 and 0.16), both short of EgoPed (2.36 and 0.33), so EgoPed-IA's weaker conditioning fidelity is a property of the interaction-sampling recipe rather than of one unlucky seed. At a matched epoch (1299) the second seed gives FID 3.320 ± 0.139, which shows that the two seeds reach their best marginal at different epochs and that per-seed checkpoint selection, not a fixed epoch, is the meaningful protocol.

D. ENCODER FREEZE VS. UNFREEZE

The most important ablation is whether to freeze the ego encoder during diffusion training. Unfrozen uses an identical architecture to EgoPed but with the ego encoder left trainable, allowing the encoder to co-adapt with the denoiser via backpropagation through the diffusion loss.

Figure 4 shows the Unfrozen training trajectory (FID and R@1 vs. epoch). Two behaviors are notable:

- **FID/R@1 tradeoff**: As training progresses, R@1 improves (encoder specializes discriminatively) but FID degrades (generation quality decreases). The two optima do *not* coincide: training-time FID is best at epoch 2099 (4.789), while R@1 continues rising long after FID has begun to deteriorate.
- **R@1 saturation far below EgoPed**: After epoch ≈3299, training-time R@1 saturates, oscillating in the 0.37–0.42 band through epoch 4999 (maximum 0.417 at epoch 4599) while FID remains between 5.1 and 6.3. Neither metric ever approaches EgoPed's operating point (FID 3.392, R@1 0.548) at any point in 5000 epochs.

Definitive evaluations (3 replications, CFG = 10) confirm the picture, and are reported in Table 11:

Unfrozen is worse than EgoPed on FID, R-Precision and MultiModality at *every* evaluated checkpoint (it is, however, slightly better on mean trajectory error, see Sec. IV-G). Later checkpoints cannot change this conclusion: even Unfrozen's best training-time R@1 in the saturation band (0.417 at epoch 4599, with training-time FID 6.339) remains far below EgoPed's definitive R@1 of 0.548. The MultiModality collapse is particularly striking: Unfrozen MM $= 2.618$ vs.

TABLE 8: Behavioral probe on the held-out split (N=1,190. GT stopping base rate 0.218). The unconditional model is behaviorally ego-blind (separation $\approx$ 0), validating the probe. EgoPed couples behavior to the ego better than the pooled baseline (higher separation, best Brier). This is a second evaluator-independent axis, after ADE/FDE, on which full-sequence conditioning wins despite its lower R@1. Unfrozen shows the MultiModality collapse in behavior space: the highest separation at the *lowest* entropy yet its Brier is no better than Pooled's (overconfidence). Bold marks the best ego-trajectory-conditioned model per column. The trained unconditional model is as ego-blind as the zeroed one (separation 0.04, Brier at chance). The oracle's 0.93 is the probe reading its own label back from the caption ("…and then stops"), which validates the probe rather than a conditioning claim.

Model	Gen. stop-rate	Separation ↑	Entropy	Brier ↓
EgoPed	0.174	0.334	0.238	0.147
EgoPed, seed B	0.216	0.325	0.264	0.167
EgoPed, seed C	0.167	0.311	0.187	0.163
Pooled	0.230	0.280	0.302	0.182
Unfrozen	0.217	0.351	0.182	0.178
EgoPed-IA (+ interaction sampling)	0.145	0.158	0.236	0.190
EgoPed-IA, epoch-matched to EgoPed (3399)	0.128	0.222	0.171	0.175
EgoPed-IA, seed 2 (best-val, epoch 899)	0.092	0.125	0.157	0.190
EgoPed-IA, seed 2 at epoch 1299	0.101	0.177	0.160	0.178
Unconditional	0.194	0.026	0.533	0.195
Unconditional (trained, p_{drop}=1)	0.232	0.045	0.590	0.195
+ interaction sampling (FID 1.43)	0.112	0.024	0.338	0.196
Vehicle-only text (epoch 3399)	0.230	0.213	0.387	0.184
Vehicle-only text, FID-best checkpoint (epoch 99)	0.101	0.049	0.301	0.189
Oracle text (told the pedestrian's behaviour)	0.243	0.936	0.023	0.030

TABLE 9: CFG sweep. EgoPed's FID is non-monotonic in guidance with an optimum at CFG 10. Pooled degrades monotonically. Pooled rows are reported under its original evaluation pipeline (the sweep predates our unified protocol); the within-model *trends*, which are the point of this table, are unaffected.

Method	CFG	FID ↓	R@1 ↑	MM ↑
Pooled	5	6.603 ±0.067	0.671 ±0.003	3.503
Pooled	10	6.963 ±0.035	0.767 ±0.006	3.031
Pooled	15	7.400 ±0.016	0.797 ±0.005	2.724
EgoPed	5	3.842 ±0.108	0.506 ±0.009	4.141
EgoPed	10	**3.392** ±0.179	0.548 ±0.002	3.956
EgoPed	15	3.547 ±0.122	0.536 ±0.003	3.910

EgoPed MM $=$ 3.956 (34% lower). This indicates near-deterministic generation per condition which shows that the ego encoder has learned highly discriminative features for each trajectory, collapsing the generated motion distribution to a narrow region.

a: Mechanism.

When the encoder is frozen (EgoPed), it maps ego trajectories to general-purpose representations aligned with the VAE latent space. The denoiser learns to query these fixed representations and generate diverse, plausible motions for each condition. When the encoder is unfrozen (Unfrozen), backpropagation through the denoising loss pushes the encoder toward representations that *minimize the denoising loss*, essentially learning discriminative features that uniquely identify each trajectory. This over-specialization constrains generation: the denoiser outputs similar samples for each unique ego representation, collapsing MultiModality.

This pattern mirrors the frozen-encoder paradigm in image generation [15]–[17]: frozen conditioning encoders preserve general-purpose representations and allow the generative model to learn diverse ways to satisfy the conditioning signal.

b: Is the pretraining necessary?

Freezing alone is not sufficient: an EgoPed model trained with a *randomly initialized* frozen ego encoder (identical otherwise) reaches only FID $= 4.968 \pm 0.171$ with R-Precision@1 exactly at chance, which is statistically indistinguishable from

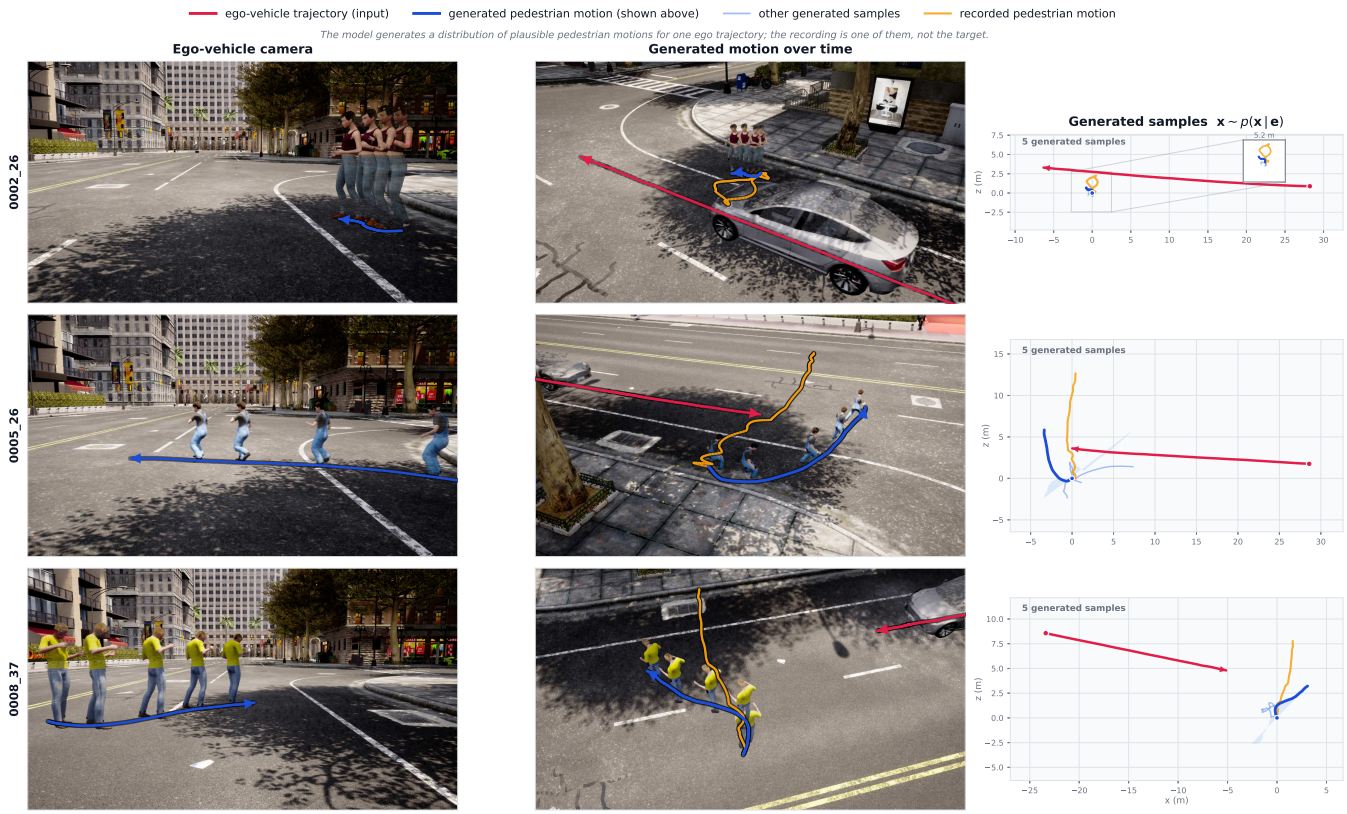


FIGURE 3: Qualitative results from EgoPed (epoch 3399, CFG=10) on three held-out ego conditions (one per row), with the generated body motion retargeted onto pedestrian characters in the CARLA simulator [75]. **Left:** the scene seen from the ego-vehicle camera. **Middle:** third-person view of the generated motion over time, overlaid with the ego-vehicle trajectory that conditions the model (red), the root trajectory of the shown sample (dark blue), and the recorded pedestrian's trajectory (orange). **Right:** bird's-eye view of five samples $\mathbf{x} \sim p(\mathbf{x} \mid \mathbf{e})$ drawn for the same ego trajectory (shown sample dark blue, the other four light blue). The model generates a distribution of plausible pedestrian motions for one ego trajectory. The recording is one outcome from that distribution, not the target. The body pose inherits the limitations quantified in Sec. IV-B: the feet slide rather than plant, in the generated motion as in the labels.

TABLE 10: Training-data pipeline 2×2 (FID ↓, unified eval, 3 replications; best checkpoint per cell). The two interventions are approximately additive: conditioning granularity dominates (≈2.7–2.8 FID in both columns), and interaction-aware sampling contributes a similar absolute gain to both architectures (−0.59 pooled, −0.51 full-sequence).

	− interaction sampling	+ interaction sampling
Pooled	6.214 ±0.105	5.620 ±0.117
EgoPed	3.392 ±0.179	**2.880** ±0.084

the ego-zeroed null branch (FID 5.18, Table 4), though that branch is itself a weak reference, and a properly trained prior reaches 3.24, so the random-encoder model is worse than simply not conditioning at all. Together with Unfrozen, this brackets the design from both sides: the alignment pretraining is what *injects* usable ego information, and freezing is what *preserves* generative diversity while using it.

E. CONDITIONING MECHANISM: SELF-ATTENTION VS. CROSS-ATTENTION

Our denoiser injects the ego sequence by concatenating it with the latent tokens and applying self-attention over the joint sequence (Section IV). A common alternative for sequence conditioning is *cross-attention*, where the latent tokens act as queries and the ego sequence as keys and values, so the ego tokens never attend among themselves and the attention cost is $\mathcal{O}(L \times T)$ rather than $\mathcal{O}((L+T)^2)$. To test whether the specific mechanism matters, we train an otherwise-identical cross-attention denoiser (same frozen VAE and frozen ego encoder, same data, same effective batch size) and evaluate it on the held-out test split.

As shown in Table 12, the self-attention and cross-attention denoisers achieve statistically comparable FID (3.338 vs. 3.490; 95% CIs $[3.15, 3.53]$ and $[3.43, 3.55]$ overlap), with

TABLE 11: Unfrozen definitive evaluation at three checkpoints. EgoPed reference included for comparison.

System	FID ↓	R@1 ↑	MM ↑	Diversity
Unfrozen, epoch 2099 (best validation FID)	4.952 ±0.088	0.263 ±0.009	2.455	5.957
Unfrozen, epoch 3299 (best R@1)	5.311 ±0.060	0.348 ±0.004	2.581	6.034
Unfrozen, epoch 3399 (matched to EgoPed)	5.079 ±0.034	0.352 ±0.009	2.618	6.064
EgoPed, epoch 3399 (frozen encoder)	**3.392** ±0.179	**0.548** ±0.002	**3.956**	5.794

TABLE 12: Conditioning-mechanism ablation. Both denoisers consume the full 196-token ego sequence and differ only in how they attend to it. Evaluated on the held-out test split (a scene-disjoint half of the validation set), CFG = 10, mean ± 95% CI over 3 replications, best checkpoint each. The two mechanisms are statistically comparable (overlapping FID intervals). Cross-attention provides no advantage. The intervals here are over sampling replications at one training seed per arm, and EgoPed's seed spread (±0.79, Sec. V-C) is far wider, so this comparison can exclude a large effect but cannot resolve a small one.

Denoiser	Attention over ego	FID ↓	R@1 ↑	MM ↑
Self-attention (EgoPed)	concatenation	**3.338** ±0.190	**0.537**	4.14
Cross-attention	queries/keys-values	3.490 ±0.059	0.501	**4.61**

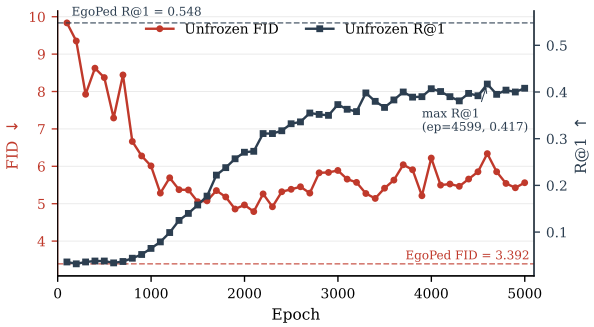


FIGURE 4: Unfrozen training trajectory: training-time validation FID (left axis, red) and R@1 (right axis, blue) vs. epoch, over the full 5000-epoch run. FID is best at epoch 2099 and degrades as R@1 rises; after epoch ≈3299, R@1 saturates in the 0.37–0.42 band while FID oscillates between 5.1 and 6.3. Dashed reference lines show EgoPed's best operating point (FID=3.392, R@1=0.548 at epoch 3399, CFG 10). Unfrozen never approaches EgoPed's performance on either metric.

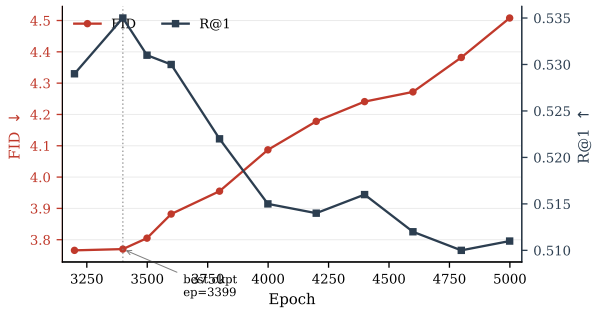


FIGURE 5: EgoPed training trajectory: training-time validation FID (left axis, red) and R@1 (right axis, blue) across the late training segment. Epoch 3399 is the best checkpoint (highest R@1, FID within noise of epoch 3199, and best definitive 3-replication FID); both metrics then degrade steadily through epoch 4999. Training loss is flat across this range — sampling quality oscillates independently of training loss.

self-attention holding a slight edge on retrieval alignment (R@1) and cross-attention generating marginally more diverse motion per condition (MM). Note that the comparison is, if anything, generous to the cross-attention variant: its best checkpoint was selected on the held-out split itself, whereas the self-attention model's checkpoint was fixed in advance from training-time validation. We also observed that the cross-attention variant trained less stably, with a noisier FID-vs-epoch trajectory.

F. CHECKPOINT SELECTION

A non-obvious finding is that FID does not monotonically improve with training. Pooled's epoch trajectory shows FID = 6.603 at epoch 4399, FID = 8.400 at epoch 4599 (regression), and FID = 7.510 at epoch 4999 (partial recovery) while displaying flat training loss throughout. EgoPed similarly peaks at epoch 3399 (definitive FID = 3.392) and degrades steadily thereafter (Fig. 5).

VI. DISCUSSION

a: Architecture recommendation.

The experimental evidence strongly favors the EgoPed architecture: frozen ego encoder + full-sequence conditioning denoiser (self-attention over the concatenated latent and ego sequence; cross-attention works equally well, Section V-E). Future work seeking to improve conditioning fidelity should explore adapter-style approaches like fine-tuning a lightweight adapter on the frozen encoder while keeping the backbone fixed rather than end-to-end unfreezing.

b: R-Precision@1 and the conditioning-diversity tradeoff.

EgoPed's R@1 (0.548) is lower than Pooled's (0.665) despite dramatically better FID. We caution against reading this as a conditioning deficit. R@1 is computed in the ego encoder's own alignment space, and each model is scored in the space its own encoder induces, so the measure is not comparable across models with differently trained encoders. The per-condition metrics of Secs. IV-G and IV-H, which use no learned embedding, rank EgoPed above Pooled on both axes. In AV simulation this is desirable: a single ego trajectory should plausibly lead to several pedestrian behaviors (crossing, waiting, accelerating). A genuine tradeoff is present too as the full-sequence conditioning generates more diverse motions per condition, which spreads the embedding distribution and lowers retrieval accuracy.

c: Broader impact.

The primary intended use of ego-conditioned pedestrian motion generation is autonomous-driving simulation. Synthesizing realistic, ego-aware pedestrian behaviors enables larger-scale safety testing without exposing real pedestrians to risk, and is particularly valuable for the long tail of rare or safety-critical interactions that are unsafe or impractical to collect. We note two risks. First, generated pedestrian motions could in principle be used adversarially, e.g. to seed motion-planning policies that learn to expect only nominal behaviors and consequently fail on out-of-distribution pedestrians. We mitigate this by reporting MultiModality and diversity metrics, which expose the breadth of the generated distribution rather than only its mean. Second, training data inherit demographic biases of the source AV datasets (AVA, nuScenes, Waymo): generated motion is conditioned on the distribution of pedestrians actually observed in U.S./European urban environments, so deployment in other settings should be preceded by additional validation. Generated motion data is synthetic and contains no personally identifiable information.

VII. LIMITATIONS AND FUTURE WORK

- **Body pose quality**: Pedestrian poses are estimated (not ground-truth 3D), and Sec. IV-B quantifies the cost: a single canonical skeleton for every pedestrian (no body-shape variation), a median foot-skate ratio of 0.97 with 94.6% of moving sequences never planting a foot, and root-relative joint variability at $0.44\times$ real mocap. However, the root trajectories are reliable while the body is a lightly articulated template sliding along them. Direct 3D body annotation from datasets like JRDB-Pose or BEDLAM, or a physics-based refit of the existing labels, would address this, and we regard it as the single largest quality constraint on the benchmark.
- **Conditioning scope**: We condition only on 2D ego trajectory. Incorporating relative pedestrian-ego geometry, ego velocity profile, or map elements could improve fidelity.
- **R-Precision is not cross-model comparable**: EgoPed scores 0.548 against Pooled's 0.665, but each model is evaluated in its own encoder's alignment space, so we do not treat this as evidence about conditioning. It is reported for completeness and the evaluator-independent metrics are used instead.
- **Physics**: Generated motions are not physically constrained (foot sliding, penetration). This is inherited as well as produced so a model that reproduces the data faithfully will reproduce the sliding. Post-hoc physics cleanup, or a contact-aware loss, could improve realism for simulation deployment.
- **Evaluator domain shift**: FID and Diversity rely on motion-feature encoders pretrained on HumanML3D, whose motion distribution differs from in-the-wild pedestrian motion. Absolute FID values should be compared only within our benchmark. The target distribution is also damped relative to mocap (Sec. IV-B), which makes the marginal easier to fit than pose realism alone would imply.
- **Single benchmark, no perceptual study**: All results are on our combined AVA/nuScenes/Waymo benchmark, and we do not yet report a human perceptual study of motion realism.

a: Future work.

The pose audit of Section IV-B defines the most consequential next step. Our de-sliding pilot shows that contact-corrected labels are representable by the motion autoencoder after fine-tuning, which scopes the remedy as a relabelling and retraining effort rather than an additional generation-time loss. Carrying it out would invalidate every distributional number reported here, since it changes the target distribution itself, and we therefore treat it as a separate undertaking rather than a revision of these results. Beyond the labels, three extensions follow directly from our findings. Richer conditioning like relative pedestrian–vehicle geometry, ego velocity profile, or map topology is a natural way to raise the information content of the condition beyond the 196×2 trajectory. A human perceptual study would supply the realism judgement that no distributional score on these labels can provide. Finally, the dissociation we document motivates a per-condition benchmark protocol for conditional motion generation in general: our trajectory-error and behavioral-

coupling probes are a minimum, and more discriminating conditioning tests are an open problem.

VIII. CONCLUSION

We introduced EgoPed, an ego-conditioned pedestrian body motion generation system based on latent diffusion with full-sequence ego conditioning. Our key finding is that exposing the denoiser to the full 196-token ego trajectory sequence dramatically outperforms single-token pooled conditioning: a 40% FID improvement (5.620 $\rightarrow$ 3.392) under a unified evaluation pipeline, with non-overlapping confidence intervals, verified on a scene-disjoint held-out split, on evaluator-independent trajectory metrics, and conservative with respect to both training-data curation and baseline seed selection. Our second contribution is an evaluation protocol that can establish such a claim. A well-trained unconditional model matches that FID, and one trained with interaction-aware sampling beats every conditional model at 1.43 while ignoring the vehicle entirely, so FID alone ranks marginals rather than conditioning. Per-condition trajectory error and the coupling of generated stopping decisions to the vehicle do detect conditioning, and on both full-sequence conditioning leads. Interaction-aware data sampling improves the marginal to FID = 2.880 (2.67–2.88 over two seeds) but not per-condition fidelity. A paired ablation reveals that frozen ego encoders are critical: unfreezing causes 34% MultiModality collapse as the encoder over-specializes to discriminative features.

These results establish a clear design recipe for ego-conditioned motion generation in AV settings: pretrain the ego encoder to regress frozen VAE latents, then freeze it during diffusion training and condition the denoiser on the full ego sequence. This recipe is the best-conditioned configuration we found across AVA, nuScenes, and Waymo, and provides a foundation for scalable, behavior-consistent pedestrian simulation.

ACKNOWLEDGMENT

This work is a result of the joint research project STADT:up (grant no. 19A22006N). The project is supported by the German Federal Ministry for Economic Affairs and Climate Action (BMWK), based on a decision of the German Bundestag. The authors are solely responsible for the content of this publication.

REFERENCES

[1] F. Camara, N. Bellotto, S. Cosar, F. Weber, D. Nathanael, M. Althoff, J. Wu, J. Ruenz, A. Dietrich, G. Markkula, A. Schieben, F. Tango, N. Merat, and C. Fox, "Pedestrian Models for Autonomous Driving Part II: High-Level Models of Human Behavior," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, pp. 5453–5472, 2021.

[2] A. Rasouli, I. Kotseruba, and J. K. Tsotsos, "Are They Going to Cross? A Benchmark Dataset and Baseline for Pedestrian Crosswalk Behavior," in *2017 IEEE International Conference on Computer Vision Workshops (ICCVW)*, 2017, pp. 206–213.

[3] A. Rasouli, I. Kotseruba, T. Kunic, and J. Tsotsos, "PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6261–6270.

[4] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human Trajectory Prediction in Crowded Spaces," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 961–971.

[5] T. Salzmann, B. Ivanovic, P. Chakravarty, and M. Pavone, *Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data*, 2020, pp. 683–700.

[6] T. Gu, G. Chen, J. Li, C. Lin, Y. Rao, J. Zhou, and J. Lu, "Stochastic Trajectory Prediction via Motion Indeterminacy Diffusion," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 17 092–17 101.

[7] X. Chen, B. Jiang, W. Liu, Z. Huang, B. Fu, T. Chen, and G. Yu, "Executing your commands via motion diffusion in latent space," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, arXiv:2212.04048.

[8] G. Tevet, S. Raab, B. Gordon, Y. Shafir, D. Cohen-Or, and A. H. Bermano, "Human motion diffusion model," in *International Conference on Learning Representations (ICLR)*, 2023, arXiv:2209.14916.

[9] M. Zhang, Z. Cai, L. Pan, F. Hong, X. Guo, L. Yang, and Z. Liu, "MotionDiffuse: Text-driven human motion generation with diffusion model," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024, arXiv:2208.15001.

[10] C. Guo, Y. Mu, M. G. Javed, S. Wang, and L. Cheng, "MoMask: Generative masked modeling of 3d human motions," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, arXiv:2312.00063.

[11] W. Mao, C. Xu, Q. Zhu, S. Chen, and Y. Wang, "Leapfrog diffusion model for stochastic trajectory prediction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, arXiv:2303.10895.

[12] L. Feng, M. Bahari, K. M. B. Amor, É. Zablocki, M. Cord, and A. Alahi, "UniTraj: A unified framework for scalable vehicle trajectory prediction," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024, arXiv:2403.15098.

[13] K. Karunratanakul, K. Preechakul, S. Suwajanakorn, and S. Tang, "Guided motion diffusion for controllable human motion synthesis," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, arXiv:2305.12577.

[14] Y. Xie, V. Jampani, L. Zhong, D. Sun, and H. Jiang, "OmniControl: Control Any Joint at Any Time for Human Motion Generation," in *International Conference on Learning Representations (ICLR)*, 2024.

[15] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, arXiv:2112.10752.

[16] H. Ye, J. Zhang, S. Liu, X. Han, and W. Yang, "IP-Adapter: Text compatible image prompt adapter for text-to-image diffusion models," *arXiv preprint arXiv:2308.06721*, 2023.

[17] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, arXiv:2302.05543.

[18] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuScenes: A Multimodal Dataset for Autonomous Driving," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 11 618–11 628.

[19] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine, V. Vasudevan, W. Han, J. Ngiam, H. Zhao, A. Timofeev, S. Ettinger, M. Krivokon, A. Gao, A. Joshi, Y. Zhang, J. Shlens, Z. Chen, and D. Anguelov, "Scalability in Perception for Autonomous Driving: Waymo Open Dataset," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2443–2451.

[20] F. Camara, N. Bellotto, S. Cosar, D. Nathanael, M. Althoff, J. Wu, J. Ruenz, A. Dietrich, and C. W. Fox, "Pedestrian Models for Autonomous Driving Part I: Low-Level Models, From Sensing to Tracking," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, pp. 6131–6151, 2021.

[21] I. Kotseruba, A. Rasouli, and J. K. Tsotsos, "Benchmark for Evaluating Pedestrian Action Prediction," in *2021 IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021, pp. 1257–1267.

[22] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese, and A. Alahi, "Social GAN: Socially Acceptable Trajectories with Generative Adversarial

Networks," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 2255–2264.

[23] A. Mohamed, K. Qian, M. Elhoseiny, and C. Claudel, "Social-STGCNN: A Social Spatio-Temporal Graph Convolutional Neural Network for Human Trajectory Prediction," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 14 412–14 420.

[24] Y. Yuan, X. Weng, Y. Ou, and K. Kitani, "AgentFormer: Agent-Aware Transformers for Socio-Temporal Multi-Agent Forecasting," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 9793–9803.

[25] K. Mangalam, H. Girase, S. Agarwal, K.-H. Lee, E. Adeli, J. Malik, and A. Gaidon, *It Is Not the Journey But the Destination: Endpoint Conditioned Trajectory Prediction*, 2020, pp. 759–776.

[26] K. Mangalam, Y. An, H. Girase, and J. Malik, "From Goals, Waypoints & Paths To Long Term Human Trajectory Forecasting," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 15 213–15 222.

[27] S. Shi, L. Jiang, D. Dai, and B. Schiele, "Motion Transformer with Global Intention Localization and Local Movement Refinement," in *Advances in Neural Information Processing Systems 35*, 2022, pp. 6531–6543.

[28] A. Rudenko, L. Palmieri, M. Herman, K. M. Kitani, D. M. Gavrila, and K. O. Arras, "Human Motion Trajectory Prediction: A Survey," *The International Journal of Robotics Research*, 2020.

[29] J. Liu, X. Mao, Y. Fang, D. Zhu, and M. Q.-H. Meng, "A Survey on Deep-Learning Approaches for Vehicle Trajectory Prediction in Autonomous Driving," in *2021 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, 2021, pp. 978–985.

[30] M. Petrovich, M. J. Black, and G. Varol, "Action-Conditioned 3D Human Motion Synthesis with Transformer VAE," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 10 965–10 975.

[31] ——, *TEMOS: Generating Diverse Human Motions from Textual Descriptions*, 2022, pp. 480–497.

[32] C. Guo, S. Zou, X. Zuo, S. Wang, W. Ji, X. Li, and L. Cheng, "Generating diverse and natural 3d human motions from text," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[33] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[34] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations (ICLR)*, 2021, arXiv:2010.02502.

[35] J. Ho and T. Salimans, "Classifier-free diffusion guidance," *arXiv preprint arXiv:2207.12598*, 2022.

[36] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," in *International Conference on Learning Representations (ICLR)*, 2014.

[37] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention Is All You Need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[38] J. Zhang, Y. Zhang, X. Cun, Y. Zhang, H. Zhao, H. Lu, X. Shen, and Y. Shan, "T2M-GPT: Generating human motion from textual descriptions with discrete representations," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, arXiv:2301.06052.

[39] B. Jiang, X. Chen, W. Liu, J. Yu, G. Yu, and T. Chen, "MotionGPT: Human Motion as a Foreign Language," in *Advances in Neural Information Processing Systems 36*, 2023, pp. 20 067–20 079.

[40] M. Zhang, X. Guo, L. Pan, Z. Cai, F. Hong, H. Li, L. Yang, and Z. Liu, "ReMoDiffuse: Retrieval-augmented motion diffusion model," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, arXiv:2304.01116.

[41] Y. Shafir, G. Tevet, R. Kapon, and A. H. Bermano, "Human Motion Diffusion as a Generative Prior," in *International Conference on Learning Representations (ICLR)*, 2024.

[42] M. Hassan, V. Choutas, D. Tzionas, and M. Black, "Resolving 3D Human Pose Ambiguities With 3D Scene Constraints," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 2282–2292.

[43] M. Hassan, D. Ceylan, R. Villegas, J. Saito, J. Yang, Y. Zhou, and M. Black, "Stochastic Scene-Aware Motion Prediction," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 11 354–11 364.

[44] Z. Wang, Y. Chen, T. Liu, Y. Zhu, W. Liang, and S. Huang, "HUMANISE: Language-Conditioned Human Motion Generation in 3D Scenes," in *Advances in Neural Information Processing Systems 35*, 2022, pp. 14 959–14 971.

[45] N. Jiang, Z. Zhang, H. Li, X. Ma, Z. Wang, Y. Chen, T. Liu, Y. Zhu, and S. Huang, "Scaling Up Dynamic Human-Scene Interaction Modeling," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 1737–1747.

[46] H. Liang, W. Zhang, W. Li, J. Yu, and L. Xu, "InterGen: Diffusion-based multi-human motion generation under complex interactions," *International Journal of Computer Vision*, 2024, arXiv:2304.05684.

[47] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, and K. Aberman, "DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, arXiv:2208.12242.

[48] B. Wilson, W. Qi, T. Agarwal, J. Lambert, J. Singh, S. Khandelwal, B. Pan, R. Kumar, A. Hartnett, J. K. Pontes, D. Ramanan, P. Carr, and J. Hays, "Argoverse 2: Next Generation Datasets for Self-Driving Perception and Forecasting," in *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

[49] R. Martín-Martín, M. Patel, H. Rezatofighi, A. Shenoi, J. Gwak, E. Frankel, A. Sadeghian, and S. Savarese, "JRDB: A Dataset and Benchmark of Egocentric Robot Visual Perception of Humans in Built Environments," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, pp. 6748–6765, 2023.

[50] Z. Chen, J. Yang, J. Huang, R. de Lutio, J. M. Esturo, B. Ivanovic, O. Litany, Z. Gojcic, S. Fidler, M. Pavone, L. Song, and Y. Wang, "Omnire: Omni urban scene reconstruction," in *The Thirteenth International Conference on Learning Representations*, 2025.

[51] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, "SMPL: A skinned multi-person linear model," *ACM Transactions on Graphics, (Proc. SIGGRAPH Asia)*, vol. 34, no. 6, pp. 248:1–248:16, Oct. 2015.

[52] F. Bogo, A. Kanazawa, C. Lassner, P. Gehler, J. Romero, and M. J. Black, *Keep It SMPL: Automatic Estimation of 3D Human Pose and Shape from a Single Image*, 2016, pp. 561–578.

[53] M. Kocabas, N. Athanasiou, and M. J. Black, "VIBE: Video Inference for Human Body Pose and Shape Estimation," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 5252–5262.

[54] J. Li, C. Xu, Z. Chen, S. Bian, L. Yang, and C. Lu, "HybrIK: A Hybrid Analytical-Neural Inverse Kinematics Solution for 3D Human Pose and Shape Estimation," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 3382–3392.

[55] S. Goel, G. Pavlakos, J. Rajasegaran, A. Kanazawa, and J. Malik, "Humans in 4D: Reconstructing and Tracking Humans with Transformers," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 14 737–14 748.

[56] V. Ye, G. Pavlakos, J. Malik, and A. Kanazawa, "Decoupling Human and Camera Motion from Videos in the Wild," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 21 222–21 232.

[57] S. Shin, J. Kim, E. Halilaj, and M. J. Black, "WHAM: Reconstructing World-Grounded Humans with Accurate 3D Motion," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 2070–2080.

[58] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. Black, "AMASS: Archive of Motion Capture As Surface Shapes," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 5441–5450.

[59] D. Rempe, L. J. Guibas, A. Hertzmann, B. Russell, R. Villegas, and J. Yang, *Contact and Human Dynamics from Monocular Video*, 2020, pp. 71–87.

[60] K. Xie, T. Wang, U. Iqbal, Y. Guo, S. Fidler, and F. Shkurti, "Physics-based Human Motion Estimation and Synthesis from Videos," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 11 512–11 521.

[61] S. Zhang, Y. Zhang, F. Bogo, M. Pollefeys, and S. Tang, "Learning Motion Priors for 4D Human Body Capture in 3D Scenes," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 11 323–11 333.

[62] Y. Yuan, J. Song, U. Iqbal, A. Vahdat, and J. Kautz, "PhysDiff: Physics-Guided Human Motion Diffusion Model," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 15 964–15 975.
[63] L. Mourot, L. Hoyet, F. L. Clerc, and P. Hellier, "UnderPressure: Deep Learning for Foot Contact Detection, Ground Reaction Force Estimation and Footskate Cleanup," *Computer Graphics Forum*, vol. 41, pp. 195–206, 2022.
[64] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
[65] M. J. Chong and D. Forsyth, "Effectively Unbiased FID and Inception Score and Where to Find Them," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 6069–6078.
[66] G. Parmar, R. Zhang, and J.-Y. Zhu, "On Aliased Resizing and Surprising Subtleties in GAN Evaluation," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11 400–11 410.
[67] T. Kynkäänniemi, T. Karras, M. Aittala, T. Aila, and J. Lehtinen, "The Role of ImageNet Classes in Fréchet Inception Distance," in *International Conference on Learning Representations (ICLR)*, 2023.
[68] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton, "Demystifying MMD GANs," in *International Conference on Learning Representations (ICLR)*, 2018.
[69] S. Jayasumana, S. Ramalingam, A. Veit, D. Glasner, A. Chakrabarti, and S. Kumar, "Rethinking FID: Towards a Better Evaluation Metric for Image Generation," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 9307–9315.
[70] T. Kynkäänniemi, T. Karras, S. Laine, J. Lehtinen, and T. Aila, "Improved Precision and Recall Metric for Assessing Generative Models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
[71] S. Barratt and R. Sharma, "A Note on the Inception Score," *arXiv preprint arXiv:1801.01973*, 2018.
[72] T. Unterthiner, S. van Steenkiste, K. Kurach, R. Marinier, M. Michalski, and S. Gelly, "Towards Accurate Generative Models of Video: A New Metric & Challenges," *arXiv preprint arXiv:1812.01717*, 2018.
[73] D. Tsai, J. S. Berrio, M. Shan, E. Nebot, and S. Worrall, "Ms3d++: Ensemble of experts for multi-source unsupervised domain adaptation in 3d object detection," *IEEE Transactions on Intelligent Vehicles*, pp. 1–16, 2024.
[74] ——, "Ms3d: Leveraging multiple detectors for unsupervised domain adaptation in 3d object detection," in *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, 2023, pp. 140–147.
[75] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 78, 2017, pp. 1–16.

Appendix A
UNFROZEN-ENCODER TRAINING TRAJECTORY

Table 13 lists the training-time validation metrics behind Figure 4, at the checkpoint evaluation points referenced in Section V-D.

PLACE PHOTO HERE

ERIK SCHÜTZ Biography and photo.

PLACE PHOTO HERE

MOHAN RAMESH Biography and photo.

PLACE PHOTO HERE

FABIAN B. FLOHR Biography and photo.

TABLE 13: Unfrozen: training-time validation metrics at selected checkpoint evaluation points (validation runs every 100 epochs; full curve in Figure 4). Training-time FID is best at epoch 2099 (4.789); R@1 rises until epoch ≈3299, then saturates in the 0.37–0.42 band (maximum 0.417 at epoch 4599) while FID remains between 5.1 and 6.3.

Epoch	Train-time FID ↓	Train-time R@1 ↑
99	9.835	0.036
499	8.373	0.038
999	6.011	0.065
1499	5.053	0.158
1999	4.967	0.271
2099	**4.789**	0.273
2499	5.385	0.332
2999	5.889	0.373
3299	5.273	0.398
3399	5.141	0.380
3499	5.415	0.367
3999	6.221	0.407
4599	6.339	**0.417**
4999	5.561	0.408